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Course: Calculus Beta

Assignment: MyLab Math - Uge 9

Researchers have shown that the number of successive dry days that occur after a rainstorm for a particular region is a random variable that is distributed exponentially with a mean of 7 days. Complete parts (a) and (b).

a. What is the probability that 11 or more successive dry days occur after a rainstorm?

Remember that an exponential distribution has a probability density function $f(x) = ae^{-ax}$ for x in $[0, \infty)$, where the expected value, or mean, of the distribution is given by $\mu = \frac{1}{a}$.

Let X be the random variable that represents the number of successive dry days. It is given that this random variable is distributed exponentially with a mean of 7 days. Substitute this value into the formula for the mean and solve for a .

$$\mu = \frac{1}{a}$$

$$7 = \frac{1}{a}$$

$$a = \frac{1}{7}$$

Substitute the value of a into the formula for the probability density function.

$$\begin{aligned} f(x) &= ae^{-ax} \\ &= \frac{1}{7}e^{-\frac{1}{7}x} \end{aligned}$$

Thus, the probability density function for the random variable that represents the number of successive dry days is $f(x) = \frac{1}{7}e^{-\frac{x}{7}}$.

To find the probability that 11 or more successive dry days occur after a rainstorm, integrate the probability density function over the correct interval. Start by finding the limits of integration.

$$P(X \geq 11) = \int_{11}^{\infty} \frac{1}{7}e^{-\frac{x}{7}} dx$$

Since this is an improper integral, evaluate $\lim_{b \rightarrow \infty} \int_{11}^b \frac{1}{7}e^{-\frac{x}{7}} dx$. Factor $\frac{1}{7}$ out of the integral, and then find the antiderivative of the resulting integrand, $e^{-\frac{x}{7}}$.

$$\frac{1}{7} \lim_{b \rightarrow \infty} \int_{11}^b e^{-\frac{x}{7}} dx = \frac{1}{7} \lim_{b \rightarrow \infty} \left(-7e^{-\frac{x}{7}} \right) \Big|_{11}^b$$

Now factor -7 out of the expression, substitute the limits of integration, and simplify.

$$\begin{aligned} \frac{1}{7} \lim_{b \rightarrow \infty} \left(-7e^{-\frac{x}{7}} \right) \Big|_{11}^b &= - \lim_{b \rightarrow \infty} \left(e^{-\frac{x}{7}} \right) \Big|_{11}^b \\ &= - \lim_{b \rightarrow \infty} \left(e^{-\frac{b}{7}} - e^{-\frac{11}{7}} \right) \\ &= - \left(0 - e^{-\frac{11}{7}} \right) \\ &\approx 0.2077 \end{aligned}$$

Thus, the probability that 11 or more successive dry days occur after a rainstorm is about 0.2077.

b. What is the probability that fewer than 4 dry days occur after a rainstorm?

To find this probability, integrate the probability density function over the correct interval. Start by finding the limits of integration.

$$P(X < 4) = \int_0^4 \frac{1}{7} e^{-\frac{x}{7}} dx$$

Factor $\frac{1}{7}$ out of the integral. Then recall that the antiderivative of the resulting

integrand, $e^{-\frac{x}{7}}$, was found in part (a) to be $-7e^{-\frac{x}{7}}$.

$$\frac{1}{7} \int_0^4 e^{-\frac{x}{7}} dx = \frac{1}{7} \left(-7e^{-\frac{x}{7}} \right) \Big|_0^4$$

Now factor -7 out of the expression, substitute the limits of integration, and simplify.

$$\begin{aligned} \frac{1}{7} \left(-7e^{-\frac{x}{7}} \right) \Big|_0^4 &= - \left(e^{-\frac{x}{7}} \right) \Big|_0^4 \\ &= - \left(e^{-\frac{4}{7}} - e^{-\frac{0}{7}} \right) \\ &= - \left(e^{-\frac{4}{7}} - 1 \right) \\ &\approx 0.4353 \end{aligned}$$

Thus, the probability that fewer than 4 dry days occur after a rainstorm is about 0.4353.